Cohort Connect

AI-Powered Placement Officer for University of Bristol

Executive Summary

Cohort Connect is an intelligent, closed-loop talent marketplace platform designed to revolutionise how students, alumni, and employers connect at the University of Bristol. Rather than fragmented email chains, static PDF brochures, and spreadsheet tracking, Cohort Connect automates and intelligently orchestrates the entire placement journey-from CV upload to job offer-using agentic AI, semantic matching, and real-time analytics.

Problem: UK universities lack a dedicated, AI-powered placement infrastructure comparable to India's placement officer ecosystem[21][22][23][25]. Students, alumni, and employers operate in silos. Employers struggle to find talent aligned with their requirements. Students lack warm introductions to opportunities. Career services teams manage placements manually through email and spreadsheets.

Solution: A unified platform where agentic AI reads CVs, understands job requirements semantically (not just keywords), and proactively connects the right people. It becomes a living, breathing talent engine that improves with every interaction.

Why now: Agentic AI frameworks (LangChain, LangGraph) are mature. Vector embeddings enable semantic job-to-candidate matching with ~95% accuracy[27]. Universities are actively investing in AI-enhanced employability tools[26][29][32]. This is a first-mover opportunity for University of Bristol to become a model for AI-powered career services across UK higher education.

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1. Platform Overview

1.1 Core Concept: Closed-Loop Talent Marketplace

Cohort Connect operates as a **closed-loop ecosystem**:

Student uploads CV → AI reads & profiles → Employer searches semantically →
Alumni vouch (agentic draft) → Student gets warm intro → Interview & offer →
University tracks outcome & learns

Every step generates data. Every data point improves the next match.

Example workflow:

- **Day 1:** Jayeetra uploads his CV (MSc Business Analytics, projects in Python + AI/ML solutions).
- **Day 2:** Goldman Sachs HR searches: "I need someone who can build AI/ML based solutions for some usecases."
- **Day 3:** Cohort Connect's semantic search finds Jayeetra (even though his CV never says "AI/ML solutions").
- **Day 4:** The system identifies a UoB alumnus, Bhattacharjee, now at Goldman Sachs, and auto-drafts a referral email.
- **Day 5:** Bhattacharjee clicks "approve", sends it to Jayeetra with a warm intro.
- **Day 6:** Jayeetra gets an interview.
- **Week 3:** Jayeetra gets an offer. Platform tracks: skill match (✓), referral source (alumni), interview-to-offer time (5 days).

This is not possible at scale without automation. A human placement officer would struggle to:

- Read 80 CVs and understand nuanced skills from each.
- Understand an employer's unstructured search query ("predictive models for risk").
- Know all alumni and their current employers.
- Draft personalised referral emails for each connection.
- Track all outcomes in real time.

Cohort Connect does all of this automatically.

1.2 Why It's Unique

Aspect	Traditional Systems	Cohort Connect
CV Processing	Student manually fills forms or uploads PDFs (no parsing).	AI reads PDFs, auto-extracts skills, projects, experience. Profiles auto-update.
Job Matching	Keyword search ("Python").	Semantic search ("build credit risk models"). Finds matches even if wording differs.
Employer Outreach	HR posts on boards; students apply cold.	HR searches by skill/role; Cohort Connect auto-suggests top 10 students + highlights alumni connections.
Referrals	Alumni manually compose emails or ignore opportunity.	System auto-drafts referral email; alumni click "approve" in 30 seconds.
Blind Hiring	Ad hoc or manual work-arounds.	One-click toggle to hide names, gender, photos. Default-fair hiring.

Analytics	Spreadsheets updated quarterly.	Real-time dashboards: "75% placed," "SQL is top skill," "alumni referral → offer rate: 45%."
University Brochure	Static PDF created manually once per year.	Auto-generated PDF: "Top 20 Talents from MSc Business Analytics 2025" updated weekly.

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2. For Students: The "Get Hired" Journey

2.1 Smart Profile Creation

Traditional approach: Student fills 50-field form or uploads PDF that goes into a filing system.

Cohort Connect approach:

1. Student uploads their CV (PDF).
2. AI agent reads the CV and extracts:
    - Skills (Python, SQL, Tableau, financial modelling, etc.)
    - Projects (with descriptions, outcomes, impact).
    - Internships & work experience (role, company, dates, achievements).
    - Education (degree, modules, GPA if willing to share).
    - Certifications and interests.
3. System builds their searchable profile automatically, with vector embeddings for semantic understanding.
4. Student reviews and tweaks if needed (e.g., "Highlight my AWS project more").

Outcome: Profiles are updated, searchable, and never buried in a folder.

2.2 AI Resume Feedback & Match Scoring

When a student wants to apply for a specific job, they paste the job description or search by role. The system:

1. **Analyzes the job description** and extracts core requirements (skills, experience, domain knowledge).
2. **Compares the student's profile** to the job using semantic embeddings.
3. **Returns a match score** with actionable feedback:

    Job: Senior Data Analyst - Credit Risk, Goldman Sachs

Your Match: 78/100 (Strong Fit)
✓ Skills aligned: Python, SQL, statistical modelling
⚠️ Gap: Financial domain knowledge
💡 Action: Highlight your "default probability" project in your cover letter
💡 Action: Mention familiarity with credit spreads (from Module 4)

4. **Student updates their CV or application** based on feedback, improving their chances from 78% to 85%.

Real-world impact:[27][30][33] Semantic matching reduces false negatives (missing qualified candidates) and false positives (unqualified matches). It increases interview-to-offer ratios.

2.3 Hireability Health Score

A simple, addictive metric displayed on the student dashboard:

Hireability Score: 68/100🟡 (Good)

Your 2-Week Goals:

1. "Add metrics to your latest project" (+7 points)
2. "Mention 3 quantifiable outcomes" (+5 points)
3. "Update skills with 'cloud computing'" (+4 points)

Students see exactly what improves their visibility to employers. It gamifies profile improvement.

2.4 One-Click Mentorship & Alumni Connections

Students can:

- Search alumni by role, company, industry: "Show me alumni in fintech roles."
- See which alumni graduated from their cohort and where they work.
- Request a 1-on-1 chat or mentorship.
- Get warm introductions to those alumni through the system.

Example:

- Student: "I'm applying to JP Morgan. Show me alumni in that company."
- System: "3 alumni from your programme are at JPMorgan. Arpita joined in 2023 as a data scientist. Would you like to request a chat?"
- Student clicks "Request Chat". System sends Arpita a personalised notification: "A current student is interested in your path to JPMorgan. Would you like to connect?"

This is far more effective than cold LinkedIn messages.

2.5 Opportunity Feed

Instead of a static job board, students see a **personalised opportunity feed**:

- Jobs matching their profile (ranked by fit).
- Alumni chats and mentorship requests.
- Referral opportunities ("Bhattacharjee at Goldman Sachs is looking for someone with your skills").
- Events and workshops related to their profile.
- Employer "talent hunts" that specifically mention students like them.

Engagement: Personalised feeds significantly increase student engagement compared to generic job boards.

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3. For Alumni & Employers: The "Find Talent" Journey

3.1 Semantic Search (The Game Changer)

Instead of searching for exact keywords, employers describe what they need in natural language:

Traditional: "Search for Python"

- Returns: all CVs with "Python" mentioned.
- Misses: students who know Python but never wrote the word (they listed "NumPy," "Scikit-learn," "Django").
- False positives: students who mention Python once but don't actually work with it.

Cohort Connect (semantic search):[27][30][36][39]

Employer search query: "I need someone who can build predictive models
for assessing credit default risk in investment portfolios.
Experience with financial data preferred."

Results (ranked by semantic similarity):

1. Jayeetra Bhattacharjee | Match: 94% | Skills: Python, statistical modelling, credit risk project, financial domain knowledge

2. James Chen | Match: 87% | Skills: Python, machine learning, time-series forecasting (similar to risk prediction)

3. Sofia Martínez | Match: 81% | Skills: R, predictive modelling, econometrics background

Each result includes:

- The student's profile.

- Why they match (which skills/projects align).
- A "View Full Profile" link.
- A "Request Student Details" button (if student consented to be searchable).

How it works technically:[27][30][33][36]

- Job descriptions and student profiles are converted into **vector embeddings** using models like Sentence Transformers.
- Embeddings capture semantic meaning: "credit risk modelling" and "default probability prediction" are close in vector space.
- Search uses **cosine similarity** to rank candidates: candidates whose embeddings are closest to the job requirement embedding rank highest.
- ~95% accuracy in matching relevant candidates compared to keyword-only systems.[27]

Outcome: Employers find qualified candidates they would have missed with keyword-only search. Students get discovered for roles they didn't explicitly list.

3.2 Talent Lists & Pipeline Management

Instead of one-off searches, employers can create and maintain **talent lists**:

"Backend Engineers - UoB 2025" (pipeline view)

```
├─ Qualified (5 students)
│  └─ Last updated: 2 days ago
│  └─ Skills: Python, SQL, DevOps, Cloud (AWS/GCP)
│
├─ Strong (8 students)
│  └─ Last updated: 1 week ago
│  └─ Skills: Python, SQL, but limited cloud experience
│
├─ Consider (12 students)
│  └─ Last updated: 3 weeks ago
│  └─ Skills: Python or strong foundational programming
```

The system auto-updates these lists as new students join the cohort or update their profiles. HR doesn't need to re-search; the pipeline refreshes automatically.

3.3 Blind Hiring Mode

Employers can toggle **blind hiring** with one click:

Standard View:
Name: Jayeetra Bhattacharjee | Gender: (inferred) | University: UoB
Skills: Python, SQL, ...

Blind Hiring View:
Candidate #4521

Skills: Python, SQL, ...
Experience: [masked, but included]
Projects: [masked names/companies, but details remain]

- Names, genders, photos hidden.
- Profiles ranked purely on skill and experience alignment.
- Studies show blind hiring improves diversity outcomes.[21][23][25][33]

University benefit: Cohort Connect enables data-driven fairness, which UK employers increasingly require for ESG and diversity commitments.

3.4 One-Click Employer Outreach

When an employer identifies a talent list, they can:

1. Select a group (e.g., "Top 10 Backend Engineers").
2. Click "Generate Outreach Campaign".
3. System auto-drafts a professional email/message:

   Subject: Exciting Opportunity - Backend Engineer @ [Company]

   Hi [Student Name],

   We've identified you as a top match for our Backend Engineer role
   based on your strong experience with Python and cloud infrastructure.

   [Personalised paragraph specific to their profile]

   Would you be interested in a quick chat? [Link to video call scheduler]

   Best,
   [Employer Name]
4. Employer reviews and sends (or tweaks and sends).
5. System tracks: Opens, clicks, responses, follow-ups.

Outcome: Employers dramatically reduce time-to-hire. Students receive warm, personalised messages instead of generic mass emails.

3.5 Skills Heatmap & Cohort Insights

Employers see a real-time skills landscape:

MSc Business Analytics 2025 - Skills Snapshot

Python: 82%
SQL: 75%
Tableau: 68%
Cloud (AWS/GCP): 35%

Financial Domain: ▓▓▓

- System tracks: Which alumni refer → which students → which offers. This data becomes invaluable for future cohorts.

3.7 Reputation & Impact Tracking (Alumni)

Alumni see their contribution to placement outcomes:

Your Impact This Year

✓ Referrals Made: 7
└─ Interview requests: 5
└─ Offers: 2

✓ Mentorship Chats: 3
✓ Career Talks (at university): 1 session, 40 students

🏆 Leaderboard Rank: #4 (Alumni Connector)

💬 Latest: "Thank you for referring me-I got the job at [Company]!"

This reinforces alumni engagement and makes giving back tangible.

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4. For University / Placement Teams: The "Management" Journey

4.1 Instant Cohort Talent Brochure

With one click, generate a professional PDF brochure:

```
┌──────────────────────────────────────┐
│ TOP 20 TALENT FROM MSc BUSINESS      │
│ ANALYTICS COHORT 2025                │
│ University of Bristol                │
│ Generated: 19 December 2025          │
└──────────────────────────────────────┘

EXECUTIVE SUMMARY
This cohort comprises 45 graduate analysts
trained in advanced data science, financial
modelling, and business intelligence...

TOP CANDIDATES BY DOMAIN:
[Finance & Risk]: Jayeetra Bhattacharjee, James Chen, Sofia Martínez
[Tech & Engineering]: Amir Patel, Elena Kowalski, ...
[Consulting]: ...
```

SKILLS SNAPSHOT:
Python (82%), SQL (75%), Tableau (68%), ...

RECRUITMENT CONTACT:
Professor [Name], Prof [Email]
Placement Team, Placement@bristol.ac.uk

Features:

- Auto-populated from student profiles.
- Includes top 20 (or customisable).
- Filterable by skill, domain, or company interest.
- Updated weekly or on demand.
- Branded with University logo.
- One click to email to employers, alumni, recruiters.

Impact: Replaces manual PowerPoint creation. Universities can send brochures weekly, keeping cohort visibility high.[21][22][23][25][28]

4.2 Live Analytics Dashboard

Real-time dashboards show:

📊 COHORT PROGRESS DASHBOARD – MSc Business Analytics 2025

Overall Placement Rate: 78% (35 / 45 students)
├─ Offers Received: 30
├─ Interviews in Pipeline: 8
├─ Applied (awaiting response): 5
└─ Not Started: 2

TIMELINE VIEW:
├─ Jan 2025: 15% (6 placed)
├─ Feb 2025: 45% (20 placed)
├─ Mar 2025: 67% (30 placed)
└─ Apr 2025: 78% (35 placed)

TOP OPPORTUNITIES:
├─ Finance (8 offers): Goldman Sachs, JP Morgan, ...
├─ Consulting (6 offers): Deloitte, BCG, ...
├─ Tech (7 offers): Google, Amazon, ...

TOP SKILLS DEMANDED:
├─ Python: 35 roles
├─ SQL: 32 roles
├─ Machine Learning: 18 roles
├─ Cloud (AWS/GCP): 12 roles

```
ALUMNI REFERRAL IMPACT:
 ├─ Referrals Made: 12
 ├─ Interviews from Referrals: 9 (75%)
 ├─ Offers from Referrals: 6 (50% of interviews)

PLACEMENT CHANNELS:
 ├─ Semantic Search (Cohort Connect): 18 offers
 ├─ Alumni Referral: 6 offers
 ├─ Cold Applications: 7 offers
 ├─ Campus Events: 3 offers
 └─ Other: 1 offer
```

Use cases:

- **Monthly reporting:** Share with faculty and senior leadership.
- **Identifying gaps:** "Cloud skills are low; should we adjust curriculum?"
- **Alumni engagement strategy:** "Alumni referrals convert at 50%; we should incentivise more."
- **Employer relationship building:** "Company X hired 2 from our cohort; we should deepen partnership."

4.3 Placement Tracking & Lifecycle Management

From upload to employment, the system tracks every stage:

```
STUDENT LIFECYCLE TRACKING

Student: Jayeetra Bhattacharjee
Timeline:
 ├─ 15 Jan: CV uploaded (profile auto-created)
 ├─ 22 Jan: Profile reviewed and approved by student
 ├─ 28 Jan: Appeared in 5 employer searches (semantic match)
 ├─ 30 Jan: Goldman Sachs HR viewed profile
 ├─ 02 Feb: Alumni referral drafted by Bhattacharjee (UGS Alumni)
 ├─ 03 Feb: Jayeetra received warm intro from Bhattacharjee
 ├─ 07 Feb: Interview scheduled with Goldman Sachs
 ├─ 14 Feb: Interview completed (feedback: "Excellent")
 ├─ 21 Feb: Offer received
 ├─ 28 Feb: Offer accepted

METRICS:
 ├─ Profile completeness: 95%
 ├─ Time to first interview: 28 days
 ├─ Time to offer: 43 days
 ├─ Match quality: 94% (semantic score)
 ├─ Referral effectiveness: Yes (from alumni)
```

Benefits:

- **Predict outcomes:** When a student has a 90%+ semantic match and alumni involvement, historically they have 65% offer rate.
- **Interventions:** Identify struggling students early ("Sara has uploaded CV but had 0 interviews in 3 weeks. Why? Is his profile weak? Are we not showcasing his well?").
- **Continuous improvement:** Which channels (semantic search vs. alumni referrals vs. cold apply) lead to fastest placements?

4.4 Feedback Loop & Curriculum Insights

Every placement feeds insights back to the programme:

CURRICULUM INSIGHTS (from Placement Data)

📈 High Demand Skills:
• Python: Requested in 78% of roles
• SQL: Requested in 72% of roles
• Financial Modelling: 45% (growing)
• Cloud (AWS/GCP): 28% (growing trend)

📉 Skills Gap:
• Students comfortable with Python but only 35% confident in cloud deployment.
• SQL skills strong, but few understand query optimization for large datasets.

💡 Recommendations:

1. Add "Cloud Fundamentals" module (AWS/GCP certification).
2. Introduce "Real-world SQL" with performance tuning.
3. Strengthen financial domain teaching (high-value skill).

DOMAIN INSIGHTS:
• Finance roles: 8 students placed (highest offer rate)
• Tech roles: 5 students placed (longest interview cycle)
• Consulting roles: 6 students placed (tied to case study prep)

Professors can use this to:

- Update curriculum to match employer demands.
- Identify where teaching is strong (students from this module placed faster).
- Pitch improvements to leadership ("Cloud skills gap costs us 15% of fintech roles").

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5. Technical Architecture

5.1 High-Level Architecture

```
┌─────────────────────────────────────────────────────────────┐
│ CLIENT LAYER │
│ ┌──────────────────┬──────────────────┬──────────────────┐ │
│ │ Student Portal │ Alumni Portal │ Employer Portal │ │
│ │ (React/Vue.js) │ (React/Vue.js) │ (React/Vue.js) │ │
│ └──────────────────┴──────────────────┴──────────────────┘ │
└─────────────────────────────────────────────────────────────┘
                              ↓
┌─────────────────────────────────────────────────────────────┐
│ API LAYER (FastAPI) │
│ ┌──────────────────┬──────────────────┬──────────────────┐ │
│ │ Auth │ Job Matching │ Analytics │ │
│ │ CV Upload │ Referral │ Brochure Gen │ │
│ │ Profile Mgmt │ Search │ Tracking │ │
│ └──────────────────┴──────────────────┴──────────────────┘ │
└─────────────────────────────────────────────────────────────┘
                              ↓
┌─────────────────────────────────────────────────────────────┐
│ AGENTIC AI LAYER │
│ (LangChain + LangGraph) │
│ │
│ ┌─────────────────────────────────────────────────────────┐ │
│ │ Agent 1: Profile Agent (CV Reading & Extraction) │ │
│ │ • Reads PDF CVs using document parsing │ │
│ │ • Extracts skills, projects, experience │ │
│ │ • Generates vector embeddings │ │
│ └─────────────────────────────────────────────────────────┘ │
│ │
│ ┌─────────────────────────────────────────────────────────┐ │
│ │ Agent 2: Job Matching Agent │ │
│ │ • Parses job descriptions │ │
│ │ • Compares using semantic embeddings │ │
│ │ • Generates match scores & explanations │ │
│ └─────────────────────────────────────────────────────────┘ │
│ │
│ ┌─────────────────────────────────────────────────────────┐ │
│ │ Agent 3: Referral Agent │ │
│ │ • Identifies relevant alumni for referrals │ │
│ │ • Drafts personalised referral emails │ │
│ │ • Tracks referral success │ │
│ └─────────────────────────────────────────────────────────┘ │
│ │
│ ┌─────────────────────────────────────────────────────────┐ │
│ │ Agent 4: Analytics Agent │ │
│ │ • Aggregates placement outcomes │ │
│ │ • Generates insights & recommendations │ │
│ │ • Produces auto-filled brochures │ │
│ └─────────────────────────────────────────────────────────┘ │
└─────────────────────────────────────────────────────────────┘
                              ↓
```

```
┌──────────────────────────────────────────────────────────────────────────┐
│ DATA LAYER │
│ ┌──────────────────────┬──────────────────────┬──────────────────────┐ │
│ │ PostgreSQL │ Vector DB │ File Storage │ │
│ │ (Relational) │ (pgvector) │ (S3 / Local) │ │
│ │ • Users │ • CVs │ • PDFs │ │
│ │ • Profiles │ • Job desc │ • Brochures │ │
│ │ • Jobs │ • Skills │ • Generated docs │ │
│ │ • Referrals │ • Embeddings │ │ │
│ │ • Outcomes │ │ │ │
│ └──────────────────────┴──────────────────────┴──────────────────────┘ │
└──────────────────────────────────────────────────────────────────────────┘
```

5.2 Tech Stack

Layer	Technology	Why
Backend	FastAPI (Python)	Fast, async, great for AI integrations. RESTful + WebSocket support.
Agent Framework	LangChain + LangGraph[26][29][32]	Modular, mature, built for agentic AI workflows. LangGraph handles multi-agent orchestration.
LLM	Google Gemini(3-pro)/ OpenAI GPT-5	State-of-the-art for understanding and generating text. Function calling for structured outputs.
Vector Database	PostgreSQL + pgvector extension	Store and query vector embeddings. All data in one relational DB for simplicity.
Embeddings Model	Sentence Transformers (all-mpnet-base-v2) or OpenAI Embeddings	Generate 768-dimensional embeddings. Capture semantic meaning.
Document Processing	PyPDF2, pydantic	Parse CV PDFs. Validate structured data.
Frontend	React or Vue.js	Modern, responsive UIs. Real-time dashboards.
Deployment	Docker + Kubernetes (or cloud-native: GCP Cloud Run / AWS ECS)	Scalable, reproducible environments. Cloud-agnostic.
Authentication	OAuth2 + JWT	Secure, university-integrated (e.g., UoB SSO).
Monitoring	OpenTelemetry + Prometheus	Track latency, errors, agent performance.

5.3 Agentic AI Workflows (LangGraph)

Example: Job Matching Agent Workflow

```
START
↓
[RECEIVE JOB DESCRIPTION]
↓
[PARSE JOB] → Extract requirements, skills, domain
↓
[GENERATE EMBEDDINGS] → Convert to 768-dim vector
↓
[VECTOR SEARCH] → Query pgvector for similar student profiles
↓
[RANK & SCORE] → Semantic similarity scores (0–100)
↓
[EXPLAIN MATCHES] → LLM generates "why this student matches"
↓
[FILTER] → Apply constraints (blind hiring, location, etc.)
↓
[RETURN RESULTS]
 ├─ Top 10 candidates
 ├─ Match scores
 ├─ Explanations
 └─ LinkedIn profiles / contact info
↓
END
```

Example: Referral Agent Workflow

```
START
↓
[ALUMNI SELECTS STUDENT & JOB]
↓
[RETRIEVE STUDENT PROFILE] → Skills, projects, achievements
↓
[RETRIEVE JOB DESCRIPTION] → Requirements, company, role
↓
[RETRIEVE ALUMNI PROFILE] → Current role, company, relevant experience
↓
[GENERATE REFERRAL DRAFT] → LLM creates personalised email
 ├─ Why alumnus thinks student is a fit
 ├─ Personalised paragraphs (not generic)
 └─ Link to student profile
↓
[ALUMNI REVIEWS & APPROVES]
↓
[SEND & TRACK]
 ├─ Record: sent time, recipient, alumni, student
 ├─ Log: opens, clicks, responses
```

```
└─ Update: referral success metric
↓
END
```

5.4 Data Models (Core)

```
User
 ├─ id (UUID)
 ├─ email
 ├─ name
 ├─ role (student | alumni | employer | admin)
 ├─ university_id (foreign key)
 └─ created_at

StudentProfile
 ├─ user_id (foreign key)
 ├─ cv_text (full PDF text)
 ├─ cv_embeddings (pgvector: 768-dim)
 ├─ extracted_skills (JSON: [Python, SQL, ...])
 ├─ projects (JSON: [{title, description, impact, embeddings}])
 ├─ experience (JSON: [{company, role, dates, achievements}])
 ├─ hireability_score (int: 0–100)
 ├─ profile_completeness (int: 0–100)
 ├─ job_matches (foreign key → JobMatch[])
 └─ updated_at

JobPosting
 ├─ id (UUID)
 ├─ title
 ├─ company
 ├─ description
 ├─ required_skills (extracted, JSON)
 ├─ description_embeddings (pgvector: 768-dim)
 ├─ salary_range (optional)
 ├─ location
 ├─ blind_hiring_enabled (boolean)
 ├─ posted_by (employer user_id)
 └─ created_at

JobMatch
 ├─ id (UUID)
 ├─ job_id (foreign key)
 ├─ student_id (foreign key)
 ├─ semantic_score (float: 0–1)
 ├─ explanation (string: why they match)
 ├─ student_viewed (boolean)
 ├─ student_applied (boolean)
 └─ created_at
```

```
Referral
 ├─ id (UUID)
 ├─ alumni_id (foreign key)
 ├─ student_id (foreign key)
 ├─ job_id (foreign key)
 ├─ email_draft (string)
 ├─ email_sent (datetime)
 ├─ email_opened (boolean)
 ├─ email_clicked (boolean)
 ├─ interview_scheduled (boolean)
 ├─ offer_received (boolean)
 └─ created_at
```

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6. Implementation Roadmap & MVP

6.1 Phase 1: MVP (Months 1–2) – Core Placement Loop

Scope: Prove the core concept with UoB MSc Business Analytics Cohort 2025.

Deliverables:

1. **Student Portal (Stripped Down)**
    - CV upload & auto-profile creation.
    - Match score feedback for jobs.
    - View job board.
    - No alumni connect (yet).
2. **Employer Portal (Stripped Down)**
    - Upload job or paste job description.
    - Semantic search on student profiles (no blind hiring yet).
    - View top 10 matches with explanations.
3. **Admin Dashboard**
    - View all students & profiles.
    - View all job postings.
    - Manual override if needed.
    - Basic analytics (placement count, average match score).
4. **Backend**
    - FastAPI endpoints for upload, search, profile.
    - PostgreSQL + pgvector for embeddings.

    - Basic LangChain agent for CV parsing & job matching.
    - OpenAI or Gemini integration.
5. **Deployment**
    - Local server or simple cloud (e.g., GCP Cloud Run).
    - Docker container for reproducibility.

Success Metrics:

- 45 students upload CVs.
- 5+ employers post jobs and use semantic search.
- Average match score explanation is useful (gather feedback).
- Platform can run with 2–3 person team (you + 1–2 developers).

6.2 Phase 2: Alumni & Referrals (Months 3–4)

Add to MVP:

1. **Alumni Portal**
    - View current cohort.
    - Search by student or skill.
    - Request student contact.
    - Auto-draft and send referral emails.
2. **Enhanced Student Portal**
    - See alumni at companies (one-click connection).
    - View referral requests.
    - Hireability health score.
3. **Referral Agent**
    - Auto-draft referral emails.
    - Referral tracking (opens, clicks, outcomes).
4. **Analytics Dashboard (Expanded)**
    - Placement timeline.
    - Alumni referral conversion rates.
    - Skills heatmap.

Success Metrics:

- 10+ alumni engage and send referrals.

- Referral → interview rate > 50%.
- 15+ students placed via referrals.

6.3 Phase 3: Scale & Polish (Months 5–6)

Additions:

1. **Blind Hiring Mode**
    - Toggle for employers.
    - Filter on skill/domain only.
2. **Cohort Brochure Generator**
    - Auto-generate PDF with top candidates.
    - Weekly updates.
3. **Advanced Analytics**
    - Placement tracking by timeline.
    - Curriculum insights.
    - Success story patterns.
4. **University Reporting Tools**
    - Dashboard for faculty.
    - Automated monthly reports.
5. **Scaling & Performance**
    - Optimize vector search for 100+ profiles.
    - Implement caching for embeddings.
    - Rate limiting for API calls.

Success Metrics:

- 80%+ placement rate (35+ students placed).
- Positive feedback from employers and alumni.
- Professor can generate and email a professional cohort brochure in 60 seconds.
- Ready to pitch to other UoB programmes.

6.4 Phase 4: Multi-University & Market (Months 7–12)

Long-term (post-MVP):

1. **Multi-Tenant Architecture**

    - Each university is a tenant.
    - Each programme is a cohort within a university.
    - Isolated data, shared infrastructure.
2. **University Branding**
    - Each university customises logo, colours, domain.
    - Managed onboarding for other universities.
3. **Enterprise Features**
    - Bulk employer imports.
    - API for job board integrations.
    - White-label option.
4. **Go-to-Market**
    - Case study: UoB MSc Business Analytics (placement rates, outcomes).
    - Target Russell Group and other research-intensive universities.
    - Freemium model: basic features free, premium analytics paid.

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7. Key Differentiators & Competitive Advantages

7.1 Why This Cannot Be Replaced by Existing Tools

Tool	Limitation	Cohort Connect Solution
LinkedIn	Alumni connections are weak. No placement tracking. HR can't do semantic search on student cohorts.	Closed-loop system designed specifically for university placements. Semantic search built in. All data (CVs, jobs, referrals, outcomes) in one place.
Generic Job Boards (Indeed, Glassdoor)	Open to anyone. No university affiliation. No alumni advantage. No agentic automation.	Closed to cohort (or university). Powered by alumni network. Agents auto-draft referrals, track success, improve matching.
Manual Placement Officer	Cannot read 80 CVs deeply. Cannot search 500+ jobs semantically. Cannot scale referrals.	Automates all of this. One officer + platform can manage 10 cohorts across 5 programmes.

Existing University Systems (e.g., Target Connect, TalentLinx)	Keyword-based matching. Manual effort. No agentic automation. Poor UX.	Semantic matching. Agentic automation. Modern, clean UX. Real-time analytics.

7.2 Unique Selling Points

1. **Agentic Automation**[26][29][32]
    - System proactively reads CVs, drafts referrals, suggests matches.
    - Not just a database or search tool; it's an active player in placements.
2. **Semantic Understanding**[27][30][33][36][39]
    - Understands that "credit risk modelling" = "default probability prediction" = "loss given default analysis".
    - ~95% accuracy in matching candidates to roles semantically.
3. **Closed-Loop Accountability**
    - Every interaction tracked: CV upload → search → match → referral → interview → offer.
    - University learns which channels (semantic, alumni, cold apply) work best.
    - Curriculum can be adjusted based on placement outcomes.
4. **Default-Fair Design**
    - Blind hiring is a built-in mode, not an afterthought.
    - Bias in matching can be audited and logged.
5. **White-Glove University Branding**
    - Auto-generated, professional brochures weekly.
    - University looks like it has a top-tier placement infrastructure (because it does now).
6. **Alumni Engagement Loop**
    - Alumni feel like super-connectors (they are, with system help).
    - Referral success is tracked and celebrated.
    - Loyalty to university increases.

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8. Financial & Strategic Case

8.1 Cost to Develop (Ballpark)

Item	Cost	Notes
Developer (Full-Stack AI)	£-	-
LLM API Credits	£500–2k	GPT-4o or Gemini for 6 months of pilot. Vectorization + generations.
Cloud Hosting (Dev/Prod)	£200–500	GCP Cloud Run, vector DB, storage.
Domain & SSL	£50	cohortconnect.bristol.ac.uk
Misc (tools, templates, etc.)	£-	
TOTAL	**~£-**	Comparable to hiring a junior placement officer for 1 year.

8.2 Business Model (Post-MVP)

For UoB (internal):

- Free to all students, alumni, employers in partnering programmes.
- Funding from careers/employability budget.
- Senior leadership sees ROI: 80%+ placement rates, improved employer relations, strong reputation.

For multi-university rollout:

- **Per-University License:** £5k–10k/year (includes 3–5 cohorts).
- **Per-Cohort Premium Analytics:** £2k/year (advanced dashboards, curriculum insights, dedicated support).
- **Employer Paid Listing** (optional): £100–500 to feature a job prominently (white-glove matching).
- **SaaS Model:** Small profit margin on hosting + API credits.

Rough unit economics:

- 10 UK universities × £7k average = £70k/year recurring.
- Cloud & ops cost: ~£10k/year.
- Gross margin: 85%+.

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9. Risk Mitigation & Governance

9.1 Data Privacy & Ethics

Concerns:

- Student CVs contain sensitive data.
- Matching algorithms could introduce bias.
- Employer search could discriminate (despite blind hiring option).

Mitigations:

1. **GDPR Compliance**
    - All data stored within UoB or UK data centre (if cloud).
    - Clear consent forms for CV usage in matching and analytics.
    - Right to delete / opt out.
2. **Algorithmic Fairness**
    - Log all matches and explanations for audit.
    - Monthly bias checks: Are certain groups consistently downranked?
    - Blind hiring toggle available to all employers (default recommended for sensitive roles).
3. **Transparency**
    - Students can see their match scores and why (explainability).
    - Employers see which students are in blind vs. named mode.
    - Regular reports to Ethics Committee if university has one.

9.2 Quality Assurance

Before each Phase launch:

- Test CV parsing on 50+ diverse résumés.
- Validate semantic search accuracy (compare AI matches to manual expert ranking).
- Collect user feedback (students, alumni, employers).
- Run security audit (API, authentication, data handling).

9.3 Rollout Strategy

Phase 1: UoB Business Analytics 2025 (controlled, small cohort).
Phase 2: Gather feedback. Improve UX and algorithms.
Phase 3: Pitch to other UoB programmes (e.g., Management, Engineering).
Phase 4: If successful, propose to Russell Group partners.

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10. Conclusion

Cohort Connect is a transformational opportunity for University of Bristol to become a leader in AI-powered student employability. It solves a real, costly problem:

- Students struggle to network and land jobs (especially outside London).
- Employers struggle to find pipeline talent aligned with their needs.
- Alumni want to give back but lack scalable infrastructure.
- Universities manage placements manually, missing data-driven insights.

By building this platform, you:

1. **Give your cohort a real competitive advantage** (warm alumni intros, semantic job matching, real-time feedback).
2. **Create a replicable, scalable model** for other universities.
3. **Demonstrate AI's practical value** (not just hype; deployed, measurable, impactful).
4. **Build something you and others can be proud of** (closing the placement gap in UK higher ed).

Next Steps:

1. **Secure buy-in** from programme director and careers service.
2. **Identify 2–3 developers** (could be final-year students, grads, or outsourced).
3. **Set a timeline** (6–12 months for full rollout).
4. **Allocate budget** (£15k–20k for dev + hosting).
5. **Start with Phase 1:** MVP with UoB Business Analytics 2025 by end of Q1 2026.

The UK placement market is ripe for disruption. Cohort Connect is your answer.

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